WORKED SOLUTION

STEP III 1998, Question 9

Mechanics • Moments of Inertia

STEP III 1998, Question 9

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/98-s3-q9

The key idea

A solid cone is a stack of thin discs whose radius grows linearly with distance from the apex. If we know the moments of inertia of a thin disc (about its axis and about a diameter), we can add up the slices with a single integral. For the second axis we need the perpendicular axis theorem (to turn the disc's axial moment into a moment about a diameter) and the parallel axis theorem (to shift that axis up to the apex). The oscillation is then a compound pendulum, whose period needs exactly the moment of inertia about the apex.

Solution

Setting up the slices

Put the apex at the origin with the axis of the cone along the z-axis, so the base is the plane $z = h$. Let ρ be the (constant) density. The cross-section at height z is a disc of radius

$$r = \frac{az}{h},$$

by similar triangles. A slice of thickness δz is a thin disc of mass $\delta m = \rho\pi r^2\,\delta z = \rho\pi\dfrac{a^2z^2}{h^2}\,\delta z$. Integrating,

$$m = \int_0^h \rho\pi\frac{a^2z^2}{h^2}\,\mathrm{d}z = \tfrac{1}{3}\rho\pi a^2 h, \qquad \text{so} \qquad \rho\pi = \frac{3m}{a^2h}.$$

The moment of inertia about the axis

A uniform disc of mass M and radius r has moment of inertia $\frac{1}{2}Mr^2$ about its axis. (Split it into rings of radius s and mass $\frac{2Ms}{r^2}\,\delta s$; each contributes s^2 times its mass, and $\int_0^r \frac{2M}{r^2}s^3\,\mathrm{d}s = \frac{1}{2}Mr^2$.)

Every slice of the cone has its centre on the axis, so the slices' contributions simply add:

$$I_{\text{axis}} = \int_0^h \tfrac{1}{2}r^2\,\rho\pi r^2\,\mathrm{d}z = \tfrac{1}{2}\rho\pi\frac{a^4}{h^4}\int_0^h z^4\,\mathrm{d}z = \tfrac{1}{10}\rho\pi a^4 h$$
$$= \frac{1}{10}\cdot\frac{3m}{a^2h}\cdot a^4h = \frac{3}{10}ma^2,$$

as required.

The moment of inertia about an axis through the apex

We use two theorems.

- **Perpendicular axis theorem.** For a lamina lying in a plane, the moment of inertia about an axis perpendicular to the plane equals the sum of the moments of inertia about any two perpendicular axes lying in the plane and meeting it at the same point.
- **Parallel axis theorem.** If a body of mass M has moment of inertia I_G about an axis through its centre of mass G, then its moment of inertia about a parallel axis a distance d away is $I_G + Md^2$.

For a thin disc, the two diameters are equivalent by symmetry, so the perpendicular axis theorem gives $2I_{\text{diam}} = \frac{1}{2}Mr^2$, that is, $I_{\text{diam}} = \frac{1}{4}Mr^2$.

Take the axis ℓ through the apex parallel to the base (say, along the x-axis). The slice at height z has its centre of mass on the z-axis, a distance z from ℓ, and its own diameter parallel to ℓ passes through that centre. By the parallel axis theorem, the slice contributes $\left(\frac{1}{4}r^2 + z^2\right)\delta m$. Hence

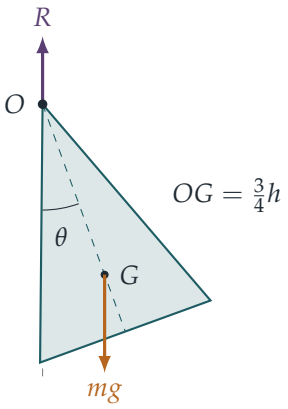
$$I_{\text{apex}} = \int_0^h \left(\frac{a^2z^2}{4h^2} + z^2\right)\rho\pi\frac{a^2z^2}{h^2}\,\mathrm{d}z = \rho\pi\frac{a^2}{h^2}\left(\frac{a^2}{4h^2}+1\right)\frac{h^5}{5}$$
$$= \frac{3m}{a^2h}\cdot\frac{a^2h^3}{5}\cdot\frac{a^2+4h^2}{4h^2} = \frac{3m}{20}\left(a^2+4h^2\right).$$

Ask yourself Why can't we apply the parallel axis theorem to the whole cone in one go, shifting from the centre of mass?

We could, but we don't yet know the moment of inertia about the parallel axis through G; that is what would need computing. Slicing avoids it, because each disc's moment about its own diameter is known, and each disc's centre is on the axis at a known distance z from ℓ. (As a by-product, $I_G = I_{\text{apex}} - m\left(\frac{3}{4}h\right)^2 = \frac{3m}{80}(4a^2+h^2)$.)

Small oscillations about the apex

Model the cone as a rigid body free to turn about a smooth horizontal axis through its apex, perpendicular to the plane of the motion. Such an axis is parallel to the base, so the moment of inertia about it is I_{apex}. Let θ be the angle between the cone's axis and the downward vertical.



The reaction at the axis has no moment about O, so the only torque is from the weight, acting at G, a distance $\frac{3}{4}h$ from O. Its moment about O is $mg \cdot \frac{3}{4}h\sin\theta$, tending to reduce θ. Taking moments about the axis,

$$I_{\text{apex}}\,\ddot{\theta} = -\tfrac{3}{4}mgh\sin\theta.$$

For small oscillations $\sin\theta \approx \theta$, so

$$\ddot{\theta} = -\omega^2\theta, \qquad \omega^2 = \frac{\frac{3}{4}mgh}{\frac{3m}{20}(a^2+4h^2)} = \frac{5gh}{4h^2+a^2}.$$

This is simple harmonic motion with period

$$\frac{2\pi}{\omega} = 2\pi\sqrt{\frac{4h^2+a^2}{5gh}},$$

as required.

Checks

- Units: $\frac{4h^2+a^2}{gh}$ has units of $\mathrm{m}^2/(\mathrm{m\,s^{-2}}\cdot\mathrm{m}) = \mathrm{s}^2$, as a period squared should.
- Thin cone ($a \to 0$): all the mass lies along the axis, with $\int z^2\,\mathrm{d}m = \frac{3}{5}mh^2$, matching $I_{\text{apex}} \to \frac{3}{5}mh^2$. The period becomes $2\pi\sqrt{4h/(5g)}$, the period of a simple pendulum of length $\frac{4}{5}h$, which is longer than $\frac{3}{4}h$, as it should be for a body whose mass is spread out along its length.
- Numerically, with $m = 1$, $a = 0.7$, $h = 2.3$, slicing with Simpson's rule and a Monte Carlo sample of points in the cone both gave $I_{\text{axis}} = 0.147$ and $I_{\text{apex}} \approx 3.2475$, matching the formulas; stepping $I\ddot{\theta} = -\frac{3}{4}mgh\sin\theta$ from a small amplitude with $g = 9.81$ gave a period of 2.7525 s, as the formula predicts.

Common slips

- Using $\frac{1}{2}r^2\,\delta m$ for each slice in the second part: about an axis parallel to the base, a disc turns about its diameter, contributing $\frac{1}{4}r^2\,\delta m$, not $\frac{1}{2}r^2\,\delta m$.
- Applying the parallel axis theorem from the apex rather than from a centre of mass: the theorem only holds when one of the two axes passes through the centre of mass.
- Treating the cone as a simple pendulum of length $\frac{3}{4}h$; the moment of inertia, not just the position of G, sets the period.

Beyond STEP

The moment of inertia about an arbitrary axis through a point is encoded in a single symmetric 3×3 matrix, the inertia tensor; the perpendicular and parallel axis theorems are special cases of how it behaves under rotation and translation. Its eigenvectors, the principal axes, explain why spinning objects are stable about some axes and wobble about others. Kleppner and Kolenkow, *An Introduction to Mechanics*, develops rigid-body rotation and the compound pendulum at this level.

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